Riyadh 4G Benchmark

Three networks compared fairly on 2.7 million phone readings



THE PROBLEM

Which of Riyadh's three 4G networks really serves people best, where they use their phones?

WHAT I BUILT

An audit of 2.7 million phone readings, a fair cell-by-cell comparison and a one-file dashboard.

THE RESULT

27.5% of the readings removed by the audit before any comparison; the fair test favours better signal, not more traffic.

27.5%
of raw readings removed by the audit, each with its rule and its reason

4.1×
how much 14 bad rows inflate one network's traffic, enough to flip the ranking

p = 0.57
10 dB more signal in the same cell and hour: 2.4% less traffic, no real effect

0.76
MASE of the SARIMA forecast, about 24% better than seasonal naive

TOOLS

Python · pandas · NumPy · statsmodels · Prophet · scikit-learn · Tableau · HTML, CSS, JavaScript

LINKS

Full project page
mellithyy.github.io/work/riyadh-4g

Live dashboard
mellithyy.github.io/riyadh-4g-network-intelligence

Code on GitHub
github.com/mellithyy/riyadh-4g-network-intelligence